Modelling Extreme Precipitation in Aotearoa (New Zealand) from 1951–2025 with Bootstrapping, Negative Binomial Regression, and Bayesian Additive Regression Trees

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Supplemental Materials

My Website

Introduction

Extreme weather threatens wildlife and public health in Aotearoa (New Zealand). Rainfall amount occurring on ‘extremely wet days’ (99th rainfall percentile) increased from 1930–2004 in Taumarunui and Queenstown (Griffiths, 2007). Rainfall extremes are predicted to increase up to 12% per Kelvin of warming in climate model simulations (Carey-Smith et al., 2010). This study aimed to answer the questions: ‣ How has extreme precipitation in Aotearoa changed over time? ‣ Is this change associated with warming temperature?

Methods

Data

Precipitation data were sourced from the Copernicus Climate Change Service (2025) ERA5 for 1951–2025, and spatially averaged to produce N=10,950 zero-dimensional (0-D) estimates. Extreme precipitation was defined by the 99th percentile (Griffiths, 2007) for the ‘baseline’ period 1951–1980 (following NASA (2025)). Annual average temperature data were sourced from NIWA’s (2025) 'Seven-station' series. Oceanic Niño Index (ONI) data were sourced from NOAA (2025).

Modelling

The Numbers of Days of Extreme Precipitation (NDEP) were compared between baseline and present (1996–2025) with a 95% prediction interval bootstrapped from the baseline data (the range of NDEP values that a future baseline-like period is expected with 95% probability to fall within). The relationship between annual NDEP and time was modelled using negative binomial regression. To model the possible association between annual NDEP and temperature, a machine learning technique (Bayesian Additive Regression Trees; BART) was used, which also controlled for the possible effect of ONI.

Results

Baseline vs Present Comparison

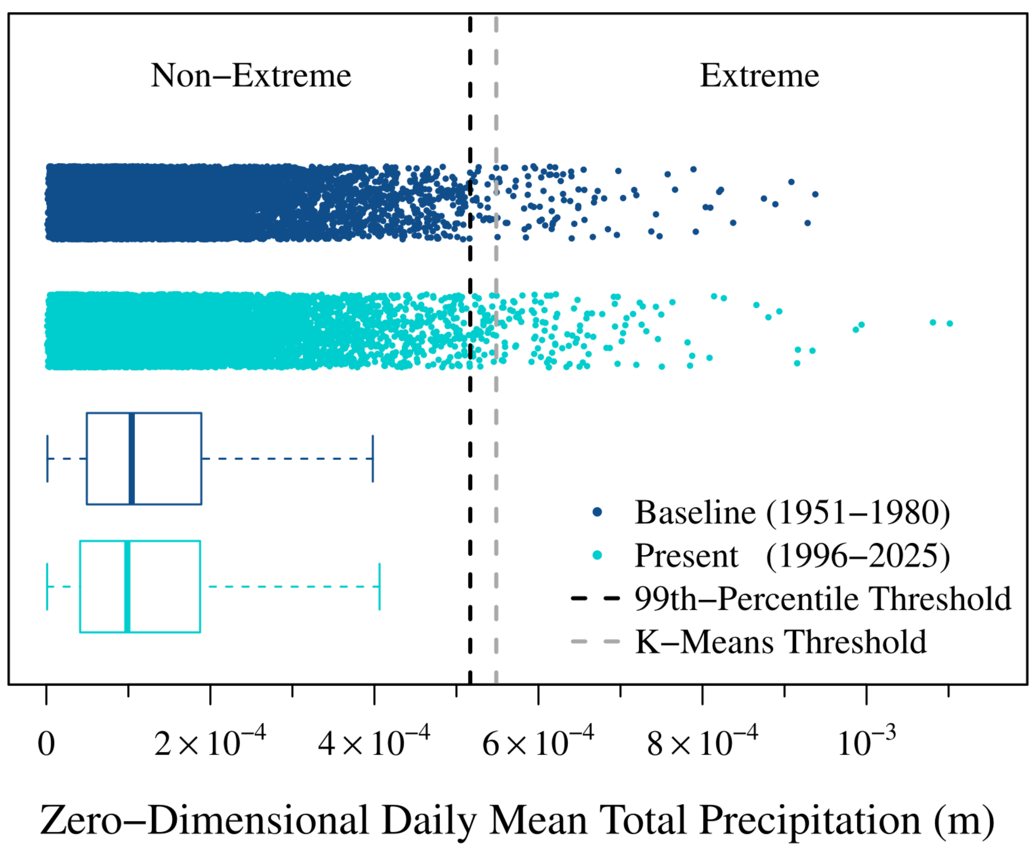
The NDEP at baseline was 110 days, with a 95% prediction interval of 90–131 days. The present NDEP was greater than this interval at 162 days. The 99th percentile threshold for extreme precipitation was similar to a k-means-based threshold.

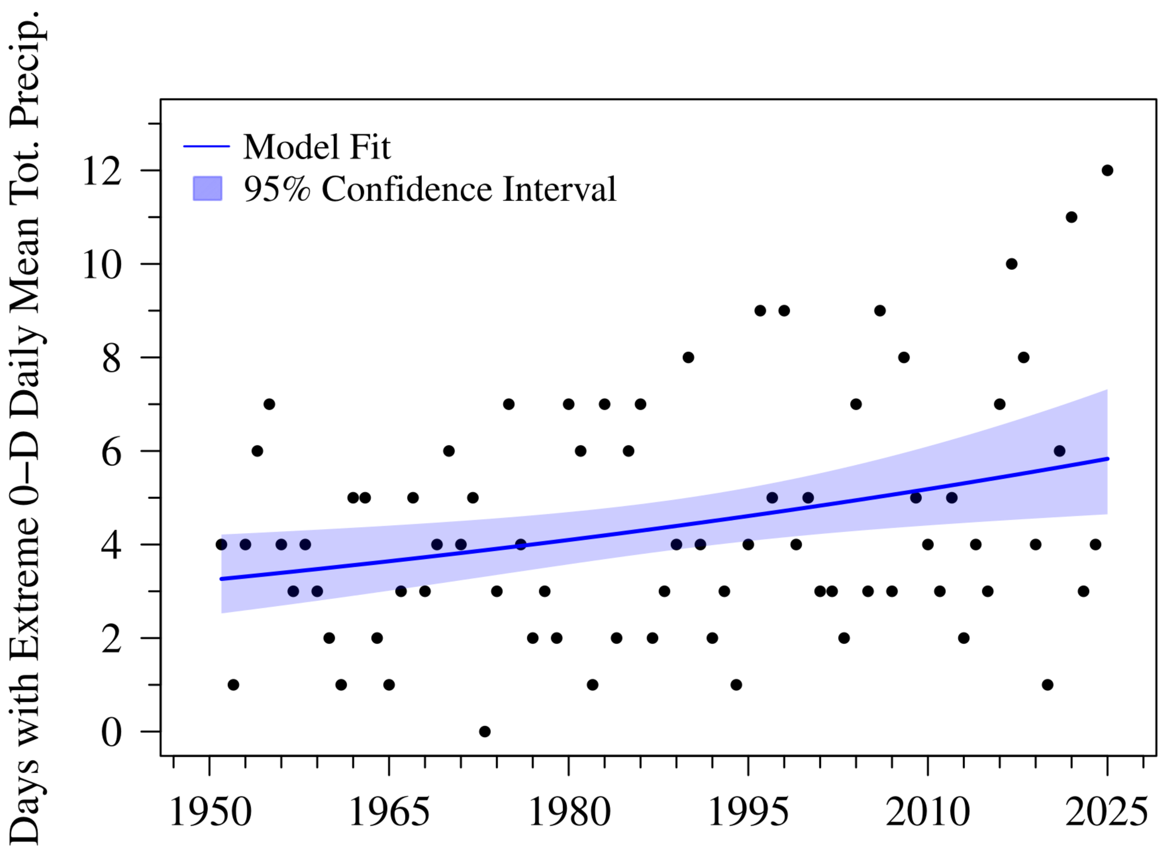
Temporal Trend

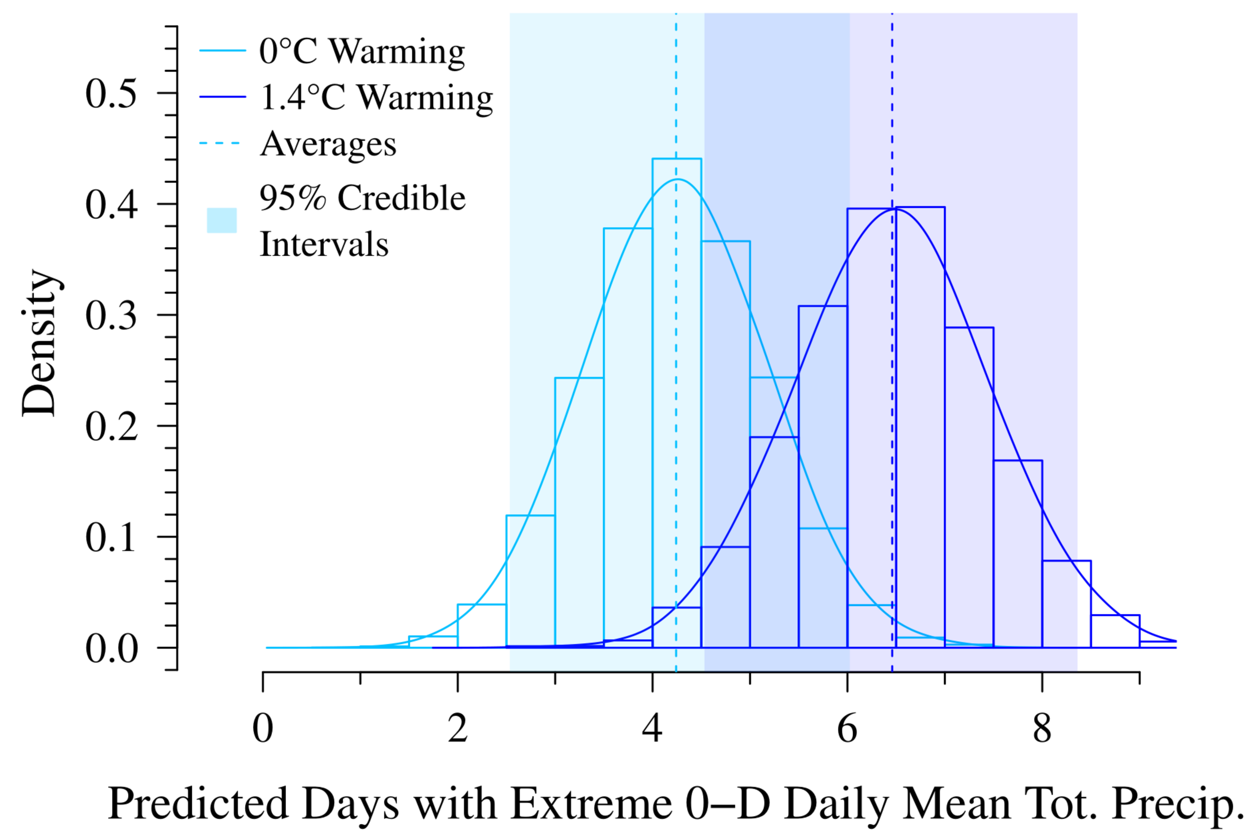
The negative binomial model performed well in diagnostic tests, had modest average error, and did not violate the assumptions underlying the model. This model suggests that, on the average, annual NDEP increased by 0.8% (95% confidence interval: 0.2–1.4%) per year from 1951–2025.

Association with Warming Temperature

BART performed well in diagnostic tests with modest average error. For fixed ONI=0, the predicted NDEP was 4.2 days (95% credible interval: 2.5–6.0 days) for 0°C of warming, and 6.5 days (4.5–8.4 days) for 1.4°C warming (the max temperature anomaly observed); NDEP increased by 37% per °C of warming, with little variation by ONI.







Discussion

This study aimed to investigate changes in extreme precipitation in Aotearoa over time and its possible association with temperature. Analyses suggest that the number of days of extreme precipitation was significantly greater in the present period compared to baseline. Indeed, on the average, the number of days of extreme precipitation appears to have increased significantly by 0.8%/year since 1951. This increase may be associated with warming; 1.4°C of warming was associated with an additional two days of extreme precipitation per year. Results were not dissimilar to the Carey-Smith et al. (2010) climate model simulation predictions of increased rainfall extremes up to 12%/K of warming (vs 37%/C in this study). Overall, these findings support and extend those of Griffiths (2007) and Carey-Smith et al. (2010) using more recent data and different modelling techniques.

Limitations

These findings should be interpreted with caution; spatially-averaged (0-D) data treat Aotearoa as a mathematical point, not allowing for spatial variation. While this provides a ‘big picture’ climate perspective, Griffiths (2007) showed significant spatial variation in Aotearoa rainfall trends. The average of drought in one region and extreme rainfall in another might result in non-extreme spatially-averaged precipitation across the two regions. There may, therefore, have been more extreme precipitation events than presented in this work. Future works may consider specialised spatiotemporal modelling techniques to investigate such variation. The modelling techniques used in this study also have limitations. For example, k-means assumes spherical clusters with equal variance and similar size. In the context of these data, the final cluster with extreme precipitation values had greater variance and smaller size than the other clusters, violating model assumptions despite resulting in a similar threshold to the 99th-percentile method. Additionally, the BART model assumed a continuous outcome, whereas the NDEP are count data. BART models for count data are novel and not yet publicly available (to this author’s knowledge); future works could implement and apply these to the ERA5 data. The BART model correctly predicted only non-negative outcome values with modest accuracy and results similar to an alternative regression model for count data (see Supplemental Materials), which suggests that these findings are not model-dependent. Still, confounding due to other factors precludes causal interpretation, and these models should not be used to forecast future climate scenarios.

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